International Journal of Image and Graphics
Vol. 12, No. 1 (2012) 1250003 (18 pages)
© World Scientific Publishing Company
DOI: 10.1142/S0219467812500039

Int. J. Image Grap. 2012.12. Downloaded from www.worldscientific.com
by 173.22.152.203 on 07/14/13. For personal use only.

AN ADAPTIVE DIFFUSION SCHEME FOR IMAGE RESTORATION AND SELECTIVE SMOOTHING

V. B. SURYA PRASATH*

Department of Mathematics, University of Coimbra
Coimbra, 3001-454, Portugal
surya@mat.uc.pt

ARINDAMA SINGH

Department of Mathematics
Indian Institute of Technology Madras
Chennai, 600036, India
asingh@iitm.ac.in

Received 3 March 2011
Revised 8 August 2011
Accepted 15 August 2011

Anisotropic partial differential equation (PDE)-based image restoration schemes employ a local edge indicator function typically based on gradients. In this paper, an alternative pixel-wise adaptive diffusion scheme is proposed. It uses a spatial function giving better edge information to the diffusion process. It avoids the over-locality problem of gradient-based schemes and preserves discontinuities coherently. The scheme satisfies scale space axioms for a multiscale diffusion scheme; and it uses a well-posed regularized total variation (TV) scheme along with Perona-Malik type functions. Median-based weight function is used to handle the impulse noise case. Numerical results show promise of such an adaptive approach on real noisy images.

Keywords: Image restoration; anisotropic diffusion; scale space; inhomogeneous PDEs; adaptive regularization.

1. Introduction

Selective smoothing of corrupted digital images is an important part of early vision. To interpret and to extract semantically meaningful objects from images we need a selective restoration method where noise and other spurious oscillations are removed and edges preserved. Real image contain diversified objects of different sizes and the imaging system is inevitably affected by noise; a multiscale analysis[1] is necessary to understand the natural images and their contents. In digital image processing,

*Corresponding author.

Int. J. Image Grap. 2012.12. Downloaded from www.worldscientific.com
by 173.22.152.203 on 07/14/13. For personal use only.

piece-wise smoothness assumption is a generic one and is used as a general tool. Advantage of such an approach is the coherence and object homogeneity present in the restored images. Therefore, selective smoothing[2,3] is an integral part of computer vision algorithms.

A closely related problem is that of detecting edges[4–8] in an image. This involves in finding edges reliably under noise and perturbations. Due to the piece-wise constant assumption of the smoothing algorithms, edges are considered to be the boundaries of such regions. Hence, edge information is automatically incorporated into adaptive smoothing and preserved or even enhanced in some cases. Linear smoothing,[2] though intuitively obvious and easy to implement, blurs the image isotropically. Nonlinear smoothing,[3] on the other hand, uses adaptive techniques by adding discontinuity measures in the scheme. These techniques are related to diffusion schemes, which are primarily of local nature. Use of partial differential equations (PDEs) for image processing is started by Perona and Malik (PM)[9] in 1990. In this methodology, adaptive smoothing is carried out using a nonuniform diffusion,[10] where important features are preserved and noise is removed. Due to its success the diffusion PDEs are widely used and studied,[11–15] and they provide a theoretical underpinning[16–18] for adaptive smoothing. PDE-based models have been applied to many machine vision problems ranging from medical images to noisy data processing.[14,17]

Among a wealth of the filters available for image denoising and restoration we mention adaptive filters,[3] convex filters,[19,20] coupled PDEs[11] and more recently trained filters paradigm.[21–23] An important aspect in anisotropic diffusion-based filters and adaptive smoothing schemes is that the feature detection is part of the problem and is built-in. For most anisotropic PDEs, this is done using the spatial gradient, giving an indication of discontinuities in the image. However, due to over-locality, it performs poorly and cannot detect significant discontinuities under noise. To overcome this, various efforts have been made in the past. These can be classified into two main approaches; one is to introduce a regularization in the diffusion PDE[16,17] and the other is to use a different feature measure[11,17,24] altogether.

Our approach is as follows. First, we introduce a new adaptive term in the diffusion PDE to effectively smooth the noisy image with control. Such a controlled diffusion acts as an adaptive scheme and uses automatic scale determination in the process. This inhomogeneity measure also constrains the diffusion spreading near edges to give better results. Various edge detection and feature detectors can be used as the inhomogeneity term in this general scheme. Second, we analyze the stability of the PDE from the scale-space perspective. Diffusion schemes are related to energy minimization;[25] thus, a connection to our adaptive diffusion PDE is indicated. Finally, various extensions are explored namely, other noise types such as salt and pepper removal and color images.

The paper is organized as follows: Sec. 2 proposes an adaptive methodology with different inhomogeneity measures and the analysis of the same is done in Sec. 3. Section 4 explores further extensions of the proposed scheme and Sec. 5

discusses simulation results. Section 6 concludes the paper indicating possible future directions in this field.

2. Proposed Inhomogeneous PDE Scheme

PM proposed[9] the following nonlinear diffusion PDE

$$\frac{\partial u}{\partial t} = \operatorname{div}(g(|\nabla u|)\nabla u) \quad \text{with } u(x,0) = I(x), \tag{1}$$

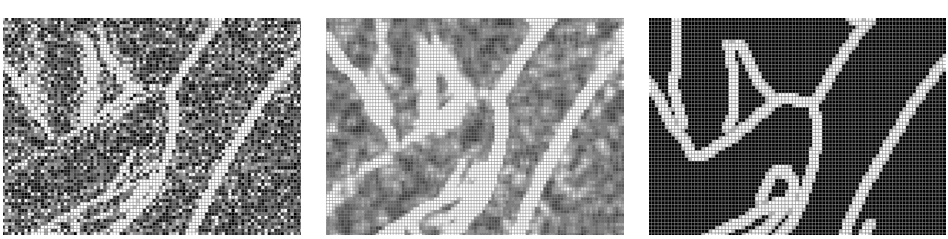
where $g : \mathbb{R}^+ \to \mathbb{R}^+$ is decreasing and is called the diffusion coefficient function and $I(x)$ denotes the noisy input image. The function g acts as a possible edge indicator in the sense that as $|\nabla u|$ increases it reduces the diffusion introducing nonlinearity in Eq. (1). Note that if $g(s) \equiv 1$, then we recover the linear diffusion. The two functions proposed in Ref. 9 are

$$g_1(s) = \exp\left(-\left(\frac{s}{K}\right)^2\right) \tag{2}$$

$$g_2(s) = \frac{1}{1 + (\frac{s}{K})^2}. \tag{3}$$

The number $K > 0$ in Eqs. (2) and (3) is known as contrast parameter.[26] It typically decides the extent of the region to be smoothed. PM equation (1) encourages intra-region smoothing than inter-region smoothing and restores the image better than the linear diffusion PDE. It creates a nonlinear scale-space where edges remain longer through larger scales and eventually converges to a constant image as $t \to \infty$. Due to good numerical results[9,14,17] coupled with the difficulty of the nonlinearity,[13] the PDE (1) has been an object of intense study by many researchers.

Under significant noise (low SNR), the gradient magnitude may not be a reliable measure as shown in Fig. 1. In Fig. 1(a), the magnitude of the gradient is shown in gray scale for visualization, where large gradient magnitude pixels are whiter and weak magnitude pixels are darker. Use of spatial gradient magnitude alone for deciding the edgeness of a pixel can give spurious results. It also leads to the loss of spatial coherence and edge continuity. A partial alleviation of the above problem



(a) (b) (c)

Fig. 1. Pixelated map showing a comparison of (a) gradient measure $|\nabla u|$, (b) smoothed gradient measure $|\nabla G_\sigma \star u|$, and (c) Canny edge detector-based smoothed measure E_ρ used in Eq. (5).

Int. J. Image Grap. 2012.12. Downloaded from www.worldscientific.com by 173.22.152.203 on 07/14/13. For personal use only.

Int. J. Image Grap. 2012.12. Downloaded from www.worldscientific.com
by 173.22.152.203 on 07/14/13. For personal use only.

is given by Catté *et al.*[16] by introducing a spatial regularization inside the PDE in Eq. (1) to get

$$\frac{\partial u}{\partial t} = \operatorname{div}(g(|\nabla G_\sigma \star u|)\nabla u), \quad \text{with } u(x,0) = I(x),$$

where $\star$ denotes convolution and $G_\sigma(x) = (2\pi\sigma^2)^{-1}\exp(-|x|^2/2\sigma^2)$ is the Gaussian kernel. This helps them to prove wellposedness of the PDE and makes the gradient robust against noise. However, using linear smoothing inside the nonlinear smoothing is against the very principle of anisotropic diffusion. In addition, due to the isotropic smoothing nature of G_σ, the usual problems associated with linear PDE such as spurious pixels leakage, no spatial coherence of edges, and most importantly, loss of localization (edges are moved from their original position) occur. Figure 1(b) shows the smoothed gradient $|\nabla G_\sigma \star u|$, compared to $|\nabla u|$ in Fig. 1(a); this has a smoother appearance but the edges are poorly localized.

It will, thus, be advantageous to use an edge indicating measure in the diffusion flow (1) along with the strength indicator $g(|\nabla u|)$. The adaptive anisotropic PDE proposed here is (we call it α-anisotropic diffusion scheme, α-ADS)

$$\frac{\partial u}{\partial t} = \operatorname{div}(\alpha(x) g(|\nabla u|)\nabla u) \quad \text{with } u(x,0) = I(x), \tag{4}$$

where $\alpha(x)$ gives a measure of inhomogeneity at every pixel x in the image domain Ω. Note that although originally PM called their PDE "anisotropic" due to its flow control measure via the nonlinearity g, it did not include any directional dependent measure. In this sense, PM equation (1) is isotropic but inhomogeneous PDE according to Weickert.[17] Our proposed adaptive anisotropic PDE (4) is an inhomogeneous PDE with a flexible adaptive term $\alpha(x)$.

Combined with $g(|\nabla u|)$, the discontinuity measure $\alpha(x)$ should give a reliable information about possible edges and boundaries using the neighborhood pixels. The general requirements for α are as follows:

(1) to reduce the amount of diffusion or to stop it at identified edge pixels; $\alpha(x) \sim 0$.
(2) in flat regions (homogeneous pixels) to let the diffusion do the work; $\alpha(x) \sim 1$.
(3) to avoid the instability, to be robust against outlier pixels, and to negotiate intermediate edge regions, $\alpha(x)$ should increase "smoothly" from 0 to 1.

Since we incorporate edge strength via the scalar diffusion function $g(|\nabla u|)$, the total coefficient $\alpha(x)g(|\nabla u|)$ in the proposed PDE (4) varies smoothly. Spatial connectivity of feature boundaries and coherency are maintained throughout the diffusion process. Various choices for the function α is discussed next.

2.1. *Edge detectors as inhomogeneity measure*

In formulating α one possible choice is to use the edge detector methods analogous to those discussed in Refs. 4–6, 8. The steps in setting up α in this case are as

Int. J. Image Grap. 2012.12. Downloaded from www.worldscientific.com
by 173.22.152.203 on 07/14/13. For personal use only.

follows:

(1) Choose an edge detector method and call the output $E(x)$, usually the output is a binary image for the edge detectors; value 1 for edge pixels and 0 for other pixels, see Fig. 1(c).
(2) Now, use Gaussian convolution to spread the edge pixels around a band of width ρ (typically we use $\rho = 1$ or 2 in our experiments, see Sec. 5). This helps to do away with outliers and to get a smoothly varying edge indicator function E_ρ:

$$E_\rho(x) = G_\rho \star E(x),$$

where Gaussian G_ρ is of width ρ. Figure (1)(c) shows the smoothed edge map E_ρ for a closeup of noisy Lena image given in Fig. 2(b). Note that, since the edge detectors typically give a coherent and continuous skeleton, the spreading function E_ρ by a Gaussian G_ρ does not move the original location of the edges detected. It only creates a smoothly varying field of width ρ around the center skeleton, avoiding the pitfalls of the method of Catté *et al.*[16]
(3) Finally, the inhomogeneity term $\alpha(x)$ is calculated using

$$\alpha(x) = \alpha_e(x) = 1 - E_\rho(x). \tag{5}$$

Assuming additive noise, one can use any of the Gaussian-based edge detectors[8] in step 1, but if the noise is of multiplicative type (impulse, salt-pepper noise) then we should use a corresponding feature detector, see Sec. 4.

Advantages of using edge detectors inside the anisotropic diffusion are as follows:

(1) Nonlinear diffusion needs to use a reliable edge indicator, as discussed gradient strength alone is not sufficient, so integrating edge finding schemes into it is advantageous.

(a) (b)

Fig. 2. Test image *Lena* 256×256 gray scale used in the experiments: (a) original and (b) Gaussian noise $\sigma = 20$ added, SNR = 8.81 dB.

(2) The mentioned edge detectors can be tuned to reflect the multiscale nature of real images,[7] features can be detected regardless of their scales.
(3) As the coupling is stronger, noise is washed away as the iteration increases and edges with lasting scales remain, giving a roadmap to the next iteration.
(4) Diffusion PDE such as Eq. (4) is known[9,16,17] to reduce noise effectively in the earlier iterations and edge detector function benefits from this and flat regions remain flat through the flow.

Thus, both the anisotropic PDE (1) and the edge indicating inhomogeneity (5) can benefit from each other as $t \to \infty$.

2.2. *Variance-based inhomogeneity measure*

One possible drawback with using the edge detectors as indicated above is the computational complexity of finding $\alpha(x)$ at every iteration t. If the noise level is moderate, one can find the edges of the initial image only and use it for all the iterations. Other choices for $\alpha(x)$ include variance controlled feature and discontinuity measures in a neighborhood of a pixel x. Motivated by the local statistical variance approach in Ref. 24, we consider a window-based estimation of image discontinuities. For example, consider the normalized variance computed in a neighborhood $\mathcal{N}_x(R)$ of the pixel $x \in \Omega$ as follows:

$$\Lambda_x(R) = \frac{\sum_{y \in \mathcal{N}_x(R)} (I(y) - \mu_x(R))^2}{|\mathcal{N}_x(R)|},$$

where $I(y)$ denotes the intensity value at pixel $y \in \Omega$ and $\mu_x(R)$ is the mean value of the pixels situated in the neighborhood $\mathcal{N}_x(R)$ of radius R. Normalize it using the minimum $\Lambda_{\min}(R)$ and maximum $\Lambda_{\max}(R)$ values found in $\mathcal{N}_x(R)$:

$$\chi_x = \frac{\Lambda_x^2(R) - \Lambda_{\min}^2(R)}{\Lambda_{\max}^2(R) - \Lambda_{\min}^2(R)}.$$

Then, use a simple function

$$\Theta(\chi_x^2(R), \theta) = \begin{cases} 0 & \chi_x^2(R) < \theta \\ \chi_x^2(R) & \chi_x^2(R) \geq \theta \end{cases}.$$

Now, set the inhomogeneity term α using the variance controlled discontinuity measure Θ by the nonlinear function (2):

$$\alpha(x) = \alpha_v(x) = \exp\left(-\Theta(\chi_x^2(R), \theta)/\delta\right). \tag{6}$$

The parameters θ and δ control the extent to which the variance induced discontinuities should be preserved. Typically, they are in the range of $\theta = 0.05$ to 0.95 and $\delta = 10$; see Sec. 5.1 for parameter selection issues.

Int. J. Image Grap. 2012.12. Downloaded from www.worldscientific.com by 173.22.152.203 on 07/14/13. For personal use only.

Int. J. Image Grap. 2012.12. Downloaded from www.worldscientific.com
by 173.22.152.203 on 07/14/13. For personal use only.

3. Analysis of the Scheme

In this section, we first relate the scheme to the energy minimization schemes. Next, a formal analysis of the inhomogeneous PDE (4) is carried out in terms of the discrete scale-space properties.

Like PM scheme (1), the PDE of our scheme (4) can considered as minimizing the energy functional in bounded variation space $BV(\Omega)$:

$$E(u) = \int_\Omega \phi(x, |\nabla u|)\, dx + \frac{1}{2}\int_\Omega |u - I|^2\, dx. \tag{7}$$

The gradient descent form of the Euler–Lagrange equation for the minimization problem gives the biased PDE,

$$\begin{aligned} \frac{\partial u}{\partial t} - \operatorname{div}(g(x, |\nabla u|)\nabla u) + (u - I) = 0 \quad &\text{on } \Omega \times [0, T], \\ u = I \quad &\text{on } \Omega \times \{0\}, \end{aligned} \tag{8}$$

where the diffusion coefficient is given by $g(x, s) = \frac{1}{s}\frac{\partial \phi(x,s)}{\partial s}$. Our scheme (4) is then a special case with $g(x, s) = \alpha(x)g(s)$. Following,[12] we see that the wellposedness of the adaptive PDE (4) depends on the convexity of the corresponding regularization function ϕ. As an example of such well-posed minimization problem, we consider the regularized total variation (TV)[27] function $\phi(s) = (\epsilon^2 + s^2)^{1/2}$ which is equivalent to setting the diffusion strength function $g(s) = (\epsilon^2 + s^2)^{-1/2}$. In our experiments, we make use of the α-adaptive regularized TV (α-ART) scheme

$$\frac{\partial u}{\partial t} = \operatorname{div}\left(\frac{\alpha(x)\nabla u}{\sqrt{\epsilon^2 + |\nabla u|^2}}\right) \tag{9}$$

$\epsilon > 0$ is a small parameter added to make the coefficient smooth at 0. For the adaptive regularization problem (7), we prove the following existence theorem

Theorem 1 (Wellposedness). *If $\alpha \geq 0$ is a continuous function on $\bar{\Omega}, \phi(x, |\nabla u|) = \alpha(x)|\nabla u|$ then (7), has a unique minimizer in $BV(\Omega) \cap L^2(\Omega)$.*

Proof. The functional in Eq. (7) is convex, coercive in $BV(\Omega) \cap L^2(\Omega)$; thus, it is enough to prove the lower semicontinuous property, i.e.,

$$E(u) \leq \liminf_{i\to\infty} E(u_k) \quad \text{if } u_k \to u \text{ in } L^1(\Omega).$$

By a standard result, Theorem 4 in Ref. 28, the energy functional (7) has a minimizer in $BV(\Omega) \cap L^2(\Omega)$. Uniqueness follows from strict convexity of the functional in Eq. (7).

To prove the lower semi-continuity, let $w \in C_0^1(\Omega)$ be such that $|w(x)| \leq \alpha(x)$ for all $x \in \Omega$, then

$$\int_\Omega u \operatorname{div} w\, dx = \lim_{k\to\infty} \int_\Omega u_k \operatorname{div} w\, dx \leq \liminf_{k\to\infty} \int_\Omega \alpha|\nabla u_k|,$$

where the inequality follows from the definition of TV of u. Now, taking supremum over w gives

$$\int_\Omega \phi(x, |\nabla u|)\, dx \le \liminf_{k\to\infty} \int_\Omega \phi(x, |\nabla u_k|)\, dx.$$

Thus, the functional E given in Eq. (7) is lower semicontinuous. □

Next, we prove that the corresponding biased PDE (8) has a unique weak solution in the following sense:

Definition 1. $u \in L^2(BV(\Omega); [0,T])$ is a weak solution of the PDE (8) if for every $v \in L^2(BV(\Omega); [0,T])$ a.e. as $\tau \in [0,T]$,

$$\int_0^\tau \int_\Omega \phi(x, |\nabla u|)\, dx\, dt - \int_0^\tau \int_\Omega (u - I)\,(v - u)\, dx\, dt \le \int_0^\tau \int_\Omega u_t\,(v - u)\, dx\, dt + \int_0^\tau \int_\Omega \phi(x, |\nabla v|)\, dx\, dt. \tag{10}$$

Theorem 2. *If $u_0 \in BV(\Omega) \cap L^\infty(\Omega)$ then there exists a unique weak solution $u \in L^\infty(BV(\Omega) \cap L^\infty(\Omega); [0,T])$ of Eq.* (8).

Proof. We consider the regularized TV functions $\phi_\epsilon(x, |\nabla u|) := \alpha(x)(\epsilon^2 + |\nabla u|^2)^{1/2}$ in the approximation problem of Eq. (8). Then, the proof depends on the following steps:

(1) The approximation problem with $g_\epsilon(x, |\nabla u|) = \alpha(x)(\epsilon^2 + |\nabla u|^2)^{-1/2}$, $I^\delta \in C^\infty(\bar{\Omega})$, $I^\delta \to I$ in $L^2(\Omega)$, and $\|I^\delta\|_{L^\infty} \le C\,\|I\|_{L^\infty}$ is given by

$$-\epsilon\Delta u + \frac{\partial u}{\partial t} - \operatorname{div}(g_\epsilon(x, |\nabla u|)\nabla u) + (u - I) = 0 \quad \text{on } \Omega \times [0,T],$$
$$u = I^\delta \quad \text{on } \Omega \times \{0\}$$

It admits a unique weak solution $u_\epsilon^\delta \in L^\infty(W^{1,2}(\Omega); [0,T])$ and is bounded in $L^\infty(\Omega)$ by the initial value I, i.e., $\|u_\epsilon^\delta\|_{L^\infty(\Omega\times[0,T])} \le C\,\|I\|_{L^\infty}$. These assertions follow from nonlinear semigroup theory, see, for example, Theorem 3.1 in Ref. 29, since $\mathcal{A}(u) = (u - I) - \operatorname{div}(g_\epsilon(x, |\nabla u|))$ is a maximal monotone operator.

(2) We let $\epsilon \to 0$, to get $u_\epsilon^\delta \to u^\delta$ in $L^2(\Omega \times [0,T])$ (up-to subsequences) for $u^\delta \in L^\infty(BV(\Omega) \cap L^\infty(\Omega); [0,T])$ as a weak solution of the approximate PDE with initial condition I^δ. This follows from the Sobolev inequality and uniform convergence of ϕ_ϵ to ϕ.

(3) Using the $L^\infty(\Omega)$ bound $\|u^\delta\|_{L^\infty(\Omega\times[0,T])} \le C\,\|I\|_{L^\infty(\Omega)}$, the definition of a weak solution (10) and passing to the limit $\delta \to 0$, $u^\delta \to u$ in $L^2(\Omega \times [0,T])$ (up-to subsequences) we get a weak solution u as required. Uniqueness follows from the weak solution inequality (10) above. □

Int. J. Image Grap. 2012.12. Downloaded from www.worldscientific.com by 173.22.152.203 on 07/14/13. For personal use only.

The wellposedness of the adaptive minimization (7) and the corresponding evolution PDE (8) for a general class of ϕ functions can be proved under some technical conditions. These and other relevant results are to be discussed elsewhere. The PM coefficients (2–3) correspond to nonconvex functions $\phi_1(s) = K^2/2 - K^2 \exp(-s^2/K^2)/2$ and $\phi_2(s) = K^2/2 \log(1 + (s/K)^2)$, respectively, and thus they are ill-posed.[13] Nevertheless, we do get scale-space axioms satisfied by these diffusion coefficients with either of the adaptive terms α_e in Eq. (5) or α_v in Eq. (6) used in our scheme (4).

The α-ADS (4) creates a continuous scale-space of images $u(x,t)$ as t increases from 0 to infinity. The scale-space properties of the continuous PDE extends to the discrete setting if certain natural axioms such as extremum principle, convergence, and average gray-level invariance are satisfied by the discrete scheme. Before we check the scale-space axioms, we write out the discretized version of the proposed PDE (4) as follows.

Let h be the grid size, U_{ij}^t is the intensity value $u(i,j)$ at iteration t. Instead of the classical explicit scheme, which severely restricts the step size, we make use of the unconditionally stable semi-implicit scheme. In 1D with matrix-vector notation, it reads as:

$$U^{t+1} = [1 - \tau A(U^t)]^{-1} U^t,$$

where τ is the time step, $A(U^t) = [a_{ij}(U^t)]$, and

$$a_{ij}(U^t) := \begin{cases} \dfrac{\gamma_i^t + \gamma_j^t}{2h^2} & j \in \mathcal{N}_i \\ -\displaystyle\sum_{k \in \mathcal{N}_i} \dfrac{\gamma_i^t + \gamma_k^t}{2h^2} & j = i \\ 0 & \text{otherwise} \end{cases},$$

with $\gamma_i = \alpha_i g_i$. For n-D images, the semi-implicit scheme is written as

$$U^{t+1} = \left[1 - \tau \sum_{l=1}^{n} A_l(U^t)\right]^{-1} U^t. \tag{11}$$

The matrix $A_l = (a_{ijl})_{ij}$ corresponds to derivatives along the lth coordinate axis.

Theorem 3. *The discrete version* (11) *of the proposed adaptive scheme* (4) *with* g *in Eq.* (2) *and with either edge detector-based inhomogeneity* α_e (5) *or variance-based inhomogeneity* α_v (6) *satisfies the discrete scale-space properties.*

Proof. The discrete scale-space properties constitute the preservation of average gray-value, maximum-minimum principle, having a large class of smoothing Lyapunov functionals, and convergence to a steady state. We rewrite the discrete

Int. J. Image Grap. 2012.12. Downloaded from www.worldscientific.com by 173.22.152.203 on 07/14/13. For personal use only.

scheme as:

$$U^0 = I,$$
$$U^{t+1} = Q(U^t)U^t, \quad \text{for all } t.$$

It is enough to check[17] the following criteria for $Q(U^t) = [q_{ij}(U^t)]$:

The matrix Q is symmetric nonnegative with positive diagonal and has unit row sum, also it is continuous in its argument and has property of irreducibility, i.e., any two pixels can be connected by a path with nonvanishing diffusivities.

Since the elements of the matrix in A depend only on the diffusion function g and the proposed α function, and $\alpha \in C^\infty(\Omega)$ with $0 \leq \alpha_i \leq 1$, we find that $\gamma_i = \alpha_i g_i$ is continuous and bounded above by 1. Thus, by Theorem 8 of Ref. 17, the scheme (11) satisfies the discrete scale-space properties. □

Theorem 3 guarantees[30] that the discrete scale-space of Eq. (4) cannot create new structures and is reliable under noise. Numerical simulations corroborate this assertion as we will see in Sec. 5.

4. Extensions

Typically, the PDE-based image restoration schemes such as Eq. (4) assume additive Gaussian noise to be present in the system. Hence, they fail when the image is corrupted by other type of noises. One particular case is *impulse noise*, which is common in faulty sensors, digital camera images, etc. To handle these multiplicative noise patterns effectively, we need a suitable adaptive indicator function. Here, our objective is to detect the noisy pixels correctly and then use that information in the PDE for restoring them.

The first step concerns a robust impulse detection scheme that must detect the corrupted pixels with accuracy. Various detection schemes exist for determining noisy pixels or their positions. Many of the methods[31] are based on median or adaptive median filters (AMF) alone. A drawback in these methods is that they can give high false detection when noisy pixels occur together as a patch. This will be mistakenly detected as an edge feature. To avoid this, we combine random noise detector based on AMF with the PDE as follows:

Step 1: (Noisy pixels identification) In this step, we use the AMF to find the noisy impulse pixels. For each pixel $x \in \Omega$, consider a neighborhood $\mathcal{N}_x$ of size $w \times w$ centered at that pixel. Initially set $w = 3$,

(1) Compute minimum $\Lambda_{\min}(w)$, maximum $\Lambda_{\max}(w)$, and median $\Lambda_{\text{med}}(w)$ values in $\mathcal{N}_x$.
(2) If $\Lambda_{\min}(w) < \Lambda_{\text{med}}(w) < \Lambda_{\max}(w)$, then go to step 4, else set $w = w + 2$.
(3) If $w \leq w_{\max}$, go to step 1, else set $\chi(x) = 1$.
(4) If $\Lambda_{\min}(w) < I(x) < \Lambda_{\max}(w)$, then $x \in \Omega$ is not a noisy candidate, else set $\chi(x) = 1$.

Int. J. Image Grap. 2012.12. Downloaded from www.worldscientific.com by 173.22.152.203 on 07/14/13. For personal use only.

The iteration can be stopped at $w_{\max} = 5$ or 7. Note that the original image I is unaltered and only noisy candidate pixel positions are noted from this step.

Step 2: (Adaptive-biased PDE) Next, the biased version of the PDE (4) is used to restore the pixels identified from the first step. The biased term is necessary as we want to penalize the distance between the given image I and parts of image, which have impulse free edges. We take

$$\frac{\partial u}{\partial t} = \operatorname{div}(\alpha(x) g(|\nabla u|)\nabla u) - (1 - \alpha(x))(u - I),$$

with

$$\alpha_i(x) = \begin{cases} 1 & \text{if } \chi(x) = 1 \\ \exp\left(-(|\nabla G_\rho \star u|/K)^2\right) & \text{otherwise.} \end{cases} \tag{12}$$

In particular, we use the well-posed ART regularization part (9) to get the biased α_i-ART scheme

$$\frac{\partial u}{\partial t} = \operatorname{div}\left(\frac{\alpha_i(x)\nabla u}{\sqrt{\epsilon^2 + |\nabla u|^2}}\right) - (1 - \alpha_i(x))(u - I), \tag{13}$$

with $u(x, 0) = I(x)$.

5. Numerical Results and Discussion

In this section, we show the versatility of the proposed scheme on various real and noisy images. A comparison with the original[9] PM equation (1), an adaptive linear scheme,[32] adaptive TV[33] and another PDE (hyperbolic) paradigm of shock filters[34] are undertaken for the α-ADS with both choices of inhomogeneity terms (α_e in Eq. (5) and α_v in Eq. (6)) and α-ART scheme (9) to show the effectiveness of such adaptive inhomogeneities in diffusion schemes. Further, α_i-ART (13) is compared with median-based filters[31] in impulse noise images.

To speed up the computations, all our schemes of the form (4) are implemented using the additive operator splitting (AOS) method:

$$U^{t+1} = \frac{1}{n}\sum_{l=1}^{n}[I - n\tau A_l(U^t)]^{-1}U^t$$

with reflecting boundary values and $n = 2$ for 2D images.

5.1. *Parameter selection*

Various methods[26] exist for selecting the gradient threshold K in the diffusion coefficient function Eq. (2) used in Eq. (4). For the α_e-ADS scheme (4) with Canny edge-detector-based E_ρ, we use the original parameter tuning suggested by Ref. 9. Since we already use Canny edge detector in our scheme for α_e, setting the threshold K accordingly is computationally simple.

Int. J. Image Grap. 2012.12. Downloaded from www.worldscientific.com by 173.22.152.203 on 07/14/13. For personal use only.

Other parameters, namely the two thresholds involved in Canny edge detector, are taken care by the default settings in MATLAB7. We found that the global threshold in Canny edge detector depends on the multi-scale nature of the edges and it influences the restoration only when noise level is high, $\sigma > 40$. In variance-based inhomogeneity α_v (6), there are two parameters θ and δ, which control extent of contextual discontinuities to be included. We found after a number of experiments that when θ varies 00.5–0.95, the results depend heavily, with larger the θ, smoother the resultant image. For the reported experiment (see Fig. 3(d)) with α_v-ADS scheme, we set $\theta = 0.5$ and $\delta = 10$. This middle value for θ works well for a variety of images and it can be made adaptive if the image is known to contain textures. For the impulse noise removal scheme α_i-ART (12–13), we use the same method of choosing K as in α_e-ADS. For the α-ART scheme (9), we use the corresponding parameter selection as indicated for α_e and α_v.

To make a fair comparison with our scheme, PM equation,[9] adaptive linear scheme,[32] adaptive TV,[33] the anisotropic shock filter,[34] and AMF schemes,[31] parameters are tuned using the signal to noise ratio:

$$SNR(u) = 10 \log_{10} \left(\frac{\sum (u - \bar{u})^2}{\sum (u - I)^2} \right) dB,$$

where $\bar{u}$ is mean value of u and I is the noisy input image. We choose a parameter set in the corresponding algorithm that maximizes $SNR(u)$.

5.2. *Test results*

Figure 2 shows *Lena* image used in our experiments. As evident from Figs. 3(e–f) and 4(a–b), edges are recovered coherently removing the noise by our scheme. Comparison with other schemes shows that the proposed α_e-ADS, α_v-ADS, and α-ART schemes give better results. Comparative SNR values support the visual assessment; Fig. 4(c) shows the corresponding SNR evolution, as iteration t increases. In particular, the experimental results conform the assertions at the end of Sec. 2.1. Figure 5(a) shows a real-world image that is used to test the α-ADS schemes. This image is taken with a 2-Mega-pixel mobile camera with unknown noise variance and to further challenge the proposed schemes we have added Gaussian noise of variance $\sigma = 5$. The results shown in Fig. 5(b) and 5(c) show that the adaptive denoising technique succeeds in removing noise while keeping all the salient edges. We test the two-step impulse noise removal scheme α_i-ART (13) on *Peppers* image Fig. 6(a) corrupted by impulse noise of level 40%. When the impulse pixels occur together as a patch, the proposed method correctly identifies it (Fig. 6(d)) and further removal and diffusion are carried out simultaneously. Media-based methods, Fig. 6(c), perform poorly and fail to detect the noise patches, as they use only the median values to replace the noisy pixels encountered.

In Fig. 7, we show a real and noisy *Corns* color image along with the gradient magnitude (Fig. 7(d)) showing influence of the detected edge pixels on scheme α-ART. The strong response of the contours given by the edge detector from all the

Int. J. Image Grap. 2012.12. Downloaded from www.worldscientific.com
by 173.22.152.203 on 07/14/13. For personal use only.

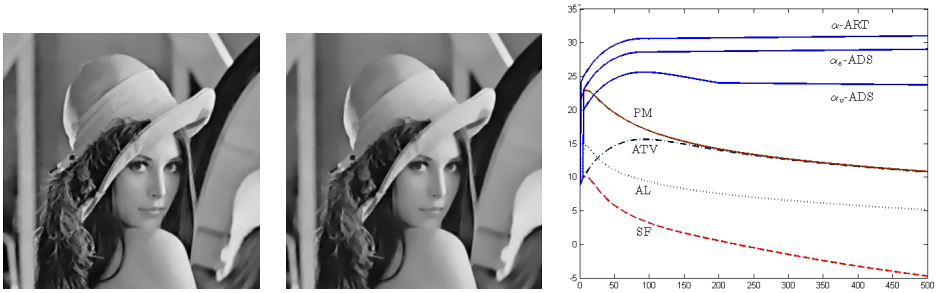
Int. J. Image Grap. 2012.12. Downloaded from www.worldscientific.com
by 173.22.152.203 on 07/14/13. For personal use only.

(a) SNR = 23.56 dB (b) SNR = 14.90 dB

(c) SNR = 15.88 dB (d) SNR = 10.85 dB

(e) SNR = 29.10 dB (f) SNR = 24.95 dB

Fig. 3. Restoration of Lena image: (a) PM equation, (b) adaptive linear (AL), (c) Adaptive TV (ATV), (d) shock filter (SF), (e) α_e-ADS scheme, and (f) α_v-ADS scheme.

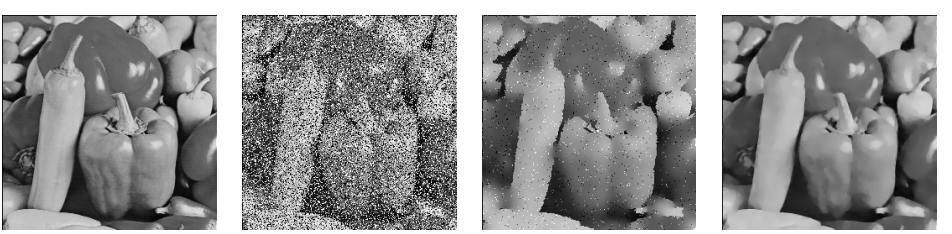


(a) SNR = 31.64 dB (b) SNR = 31.66 dB (c) Time Vs SNR

Fig. 4. α-ART scheme result of Fig. 2(b) using Canny edge detector: (a) after $t = 100$ iterations, (b) after $t = 300$ iterations, and (c) comparison of SNR evolution for the *Lena* image of all the schemes against the number of iterations.

(a) (b) SNR = 22.12 dB (c) SNR = 21.56 dB

Fig. 5. Restoration of a real image *boy* by α-ADS schemes: (a) input image captured by a mobile camera corrupted by sensor noise as well as Gaussian noise $\sigma = 5$, (b) restored result of α-ADS scheme with α_e chosen using Canny edge detector (5), and (c) restored result of α-ADS scheme with α_v chosen using the local variance-based approach (6).



(a) (b) SNR = 2.48 dB (c) SNR = 7.22 dB (d) SNR = 16.26 dB

Fig. 6. Restoration of *Peppers* image with biased α_i-ART scheme (13): (a) original, (b) 40% impulse noise corrupted image, (c) AMF-filtered image, and (d) restored by the proposed two-step scheme with 106 iterations.

Int. J. Image Grap. 2012.12. Downloaded from www.worldscientific.com by 173.22.152.203 on 07/14/13. For personal use only.

(a) (b) (c) (d)

Fig. 7. Restoration of *Corns* color image with α-ART scheme: (a) original, (b) noisy ($\sigma = 40$), (c) restored by the proposed (9) with 85 iterations, $\epsilon = 10^{-4}$, and (d) weighted edge map $\alpha(x)g(|\nabla u(x)|)$ of (c) computed from the three color channels using Canny edge detector.

Int. J. Image Grap. 2012.12. Downloaded from www.worldscientific.com by 173.22.152.203 on 07/14/13. For personal use only.

three color planes reduces the diffusion around them, thus overcoming the leakage problem of other diffusion PDE schemes.

Note that α_e-ADS and α-ART schemes give better results than α_v-ADS, in general, since α_e-ADS uses a multiscale edge detector and α-ART has strong edge preserving properties thanks to the well-posed regularized TV scheme[27] involved. Stopping time has a strong influence of the output results for the schemes in Refs. 9, 32–34 as seen from the SNR evolution curves in Fig. 4(c). Due to the presence of the term $\alpha(x)$ in the diffusion flow (4), edges are not merged as $t \to \infty$; it may be seen from the resultant images (Figs. 4(a) and 4(b)) for the α-ART scheme and the resultant SNR curve. Thus, our scheme avoids the termination problem associated with mentioned iterative diffusion PDEs. Moreover, α-ART scheme strongly smooths the regions separated by contours and gives a piece-wise constant image (see Fig. 4(b) corresponding to 300 iterations). This can be used as preprocessing step in the subsequent image-processing tasks. All the diffusion-based schemes reach optimal SNR in less number of iterations and start to reduce as iteration t increases. Our scheme (4) behaves in a similar way in the starting stages like PM equation (1) and then sustains the optimal reach using the inhomogeneity measures proposed here. PM equation (1) outperforms α_v-ADS scheme but eventually loses details in the image unlike α_v-ADS as t increases beyond 50. Finding an optimal stopping time is an important issue in time-dependent PDE-based schemes.[35] Increasing the time t reduces the small scale edge preservation in our scheme though SNR remains stable, as shown in Fig. 4(c).

We also extend the PDE scheme (4) to *color images* in a straightforward way. See Fig. 7 for a numerical example. Figure 7(d) shows all the three channel edges computed using Canny edge detector combines to give the resultant image Fig. 7(c) a natural look.

6. Conclusion

A novel adaptive anisotropic diffusion scheme is proposed based on pixel-wise estimation. This facilitates a well-defined algorithm that is flexible and general enough

to accommodate various extensions. By adding an inhomogeneity measure to the well-known Perona-Malik type PDE, we overcome the limitations present in local nature of diffusion schemes. Two inhomogeneity measures namely edge-based and variance-based are proposed to remove additive noise in digital images. To handle impulse noise effectively AMF-based inhomogeneity is used. We prove that the discrete scale-space created by the proposed adaptive PDE scheme is reliable and efficient. We also use a well-posed adaptive TV PDE in our experiments to remove noise while keeping notable discontinuities. Comparison with other schemes shows that our scheme performs well under noise. We conclude by noting that small scale textures are lost in our scheme results and hence there is a need to incorporate texture details into the flow. This may be done via Gabor type filters or other small-scale feature detectors. In addition, the parameters in α_v-ADS scheme can be made adaptive; other edge detectors[6,8] can be used in α_e as well. Apart from this, designing a robust detector α for removing a mixture of both Gaussian and impulse noises can be considered by incorporating the noises process into the PDE. These provide a future direction in adaptive diffusion PDE-based methods.

Int. J. Image Grap. 2012.12. Downloaded from www.worldscientific.com
by 173.22.152.203 on 07/14/13. For personal use only.

Acknowledgment

The research of the first author was supported by a fellowship through CMUC/FCT and also partially supported by the research project UTAustin/MAT/0009/2008 of the UT Austin | Portugal Program (`http://www.utaustinportugal.org`).

References

1. D. Marr, *Vision: A Computational Investigation into the Human Representation and Processing of Visual Information* (W. H. Freeman, San Francisco, USA, 1982).
2. J. J. Koenderink, "The structure of images," *Biological Cybernatics* **50**(5), 363–370 (1984).
3. P. Saint-Marc, J. S. Chen and G. Medioni, "Adaptive smoothing: a general tool for early vision," *IEEE Transactions on Pattern Analysis and Machine Intelligence* **13**(6), 514–529 (1991).
4. D. Marr and E. Hildreth, "Theory of edge detection," *Proceedings of the Royal Society of London B* **207**(1167), 187–217 (1980).
5. J. Canny, "A computational approach to edge detection," *IEEE Transactions on Pattern Analysis and Machine Intelligence* **8**(6), 679–698 (1986).
6. M. Heath, S. Sarkar, T. Sanocki and K. Bowyer, "Comparison of edge detectors: A methodology and initial study," *Computer Vision and Image Understanding* **69**(1), 38–54 (1998).
7. J. H. Elder, "Are edges incomplete?" *International Journal of Computer Vision* **34**(2/3), 97–122 (1999).
8. M. Basu, "Gaussian based edge-detection methods — A survey," *IEEE Transactions on Systems, Man and Cybernetics — Part C: Applications and Reviews* **32**(3), 252–260 (2002).
9. P. Perona and J. Malik, "Scale-space and edge detection using anisotropic diffusion," *IEEE Transactions on Pattern Analysis and Machine Intelligence* **12**(7), 629–639 (1990).

10. R. T. Whitaker and S. M. Pizer, "A multi-scale approach to nonuniform diffusion," *CVGIP: Image Understanding* **57**(1), 99–110 (1993).
11. M. Nitzberg and T. Shiota, "Nonlinear image filtering with edge and corner enhancement," *IEEE Transactions on Pattern Analysis and Machine Intelligence* **14**(8), 826–833 (1992).
12. Y. L. You, W. Xu, A. Tannenbaum and M. Kaveh, "Behavioral analysis of anisotropic diffusion in image processing," *IEEE Transactions on Image Processing* **5**(11), 1539–1553 (1996).
13. S. Kichenassamy, "The Perona-Malik paradox," *SIAM Journal on Applied Mathematics* **57**(5), 1328–1342 (1997).
14. G. Aubert and P. Kornprobst, *Mathematical Problems in Image Processing: Partial Differential Equation and Calculus of Variations*, 2nd edn., Vol. 147 of Applied Mathematical Sciences (Springer-Verlag, New York, USA, 2006).
15. S. Kichenassamy, "The Perona-Malik method as an edge pruning algorithm," *Journal of Mathematical Imaging and Vision* **30**(2), 209–219 (2008).
16. V. Catté, P. L. Lions, J. M. Morel and T. Coll, "Image selective smoothing and edge detection by nonlinear diffusion," *SIAM Journal on Numerical Analysis* **29**(1), 182–193 (1992).
17. J. Weickert, *Anisotropic Diffusion in Image Processing* (ECMI Series, Teubner-Verlag, Stuttgart, Germany, 1998).
18. D. Barash, "A fundamental relationship between bilateral filtering, adaptive smoothing, and the nonlinear diffusion equation," *IEEE Transactions on Pattern Analysis and Machine Intelligence* **24**(6), 844–847 (2002).
19. S. L. Keeling, "Total variation based convex filters for medical imaging," *Applied Mathematics and Computation* **139**(1), 101–119 (2003).
20. V. B. S. Prasath and A. Singh, "A hybrid convex variational model for image restoration," *Applied Mathematics and Computation* **215**(10), 3655–3664 (2010).
21. L. Shao, H. Zhang and G. de Haan, "An overview and performance evaluation of classification based least squares trained filters," *IEEE Transactions on Image Processing* **17**(10), 1772–1782 (2008).
22. L. Shao, J. Wang, I. Kirenko and G. de Haan, "Quality adaptive least squares filters for compression artifacts removal using a no-reference block visibility metric," *Journal of Visual Communication and Image Representation* **22**(1), 23–32 (2011).
23. L. Shao, H. Zhang, L. Wang and L. Wang, "Repairing imperfect video enhancement algorithms using classification-based trained filters," *Signal, Image and Video Processing*, doi: 10.1007/s11760-010-0202-8.
24. G. W. Wei, "Generalized Perona-Malik equation for image restoration," *IEEE Signal Processing Letters* **6**(7), 165–167 (1999).
25. K. N. Nordström, "Biased anisotropic diffusion: A unified regularization and diffusion approach to edge detection," *Image and Vision Computing* **8**(4), 318–327 (1990).
26. F. Voci, S. Eiho, N. Sugimoto and H. Sekiguchi, "Estimating the gradient threshold in the Perona-Malik equation," *IEEE Signal Processing Magazine* **21**(3), 39–46 (2004).
27. L. Rudin, S. Osher and E. Fatemi, "Nonlinear total variation based noise removal algorithms," *Physica D* **60**, 259–268 (1992).
28. F. Giusti, *Minimal Surfaces and Functions of Bounded Variation*, Monographs in Mathematics, Vol. 80 (Birkhaüser, Basel, Switzerland, 1984).
29. H. Brezis, *Opérateurs Maximaux Monotones et Semi-groupes de Contractions dans les Espaces de Hilbert* (North-Holland Publishing Company, Amsterdam, the Netherlands, 1973).

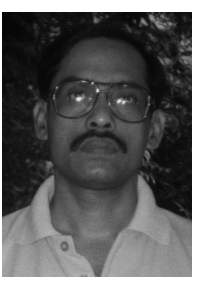
Int. J. Image Grap. 2012.12. Downloaded from www.worldscientific.com
by 173.22.152.203 on 07/14/13. For personal use only.

30. J. Weickert, B. M. H. Romeny and M. A. Viergever, "Efficient and reliable schemes for nonlinear diffusion filtering," *IEEE Transactions on Image Processing* **7**(3), 398–410 (1998).
31. H. Hwang and R. A. Haddad, "Adaptive median filters: new algorithms and results," *IEEE Transactions on Image Processing* **4**(4), 499–502 (1995).
32. W. Kusnezow, W. Horn and R. P. Würtz, "Fast image processing with constraints by solving linear PDEs," *Electronic Letters on Computer Vision and Image Analysis* **6**(2), 22–35 (2007).
33. D. M. Strong and T. F. Chan, "*Spatially and Scale Adaptive Total Variation Based Regularization and Anisotropic Diffusion in Image Processing*," Technical Report, University of California, Los Angeles, CA, CAM Report: 96–46, 1998.
34. L. Alvarez and L. Mazorra, "Signal and image restoration using shock filters and anisotropic diffusion," *SIAM Journal on Numerical Analysis* **31**(2), 590–605 (1994).
35. P. Mrázek and M. Navara, "Selection of optimal stopping time for nonlinear diffusion filtering," *International Journal of Computer Vision* **52**(2–3), 189–203 (2003).

Int. J. Image Grap. 2012.12. Downloaded from www.worldscientific.com by 173.22.152.203 on 07/14/13. For personal use only.

V. B. Surya Prasath received his Ph.D. degree in mathematics from Indian Institute of Technology Madras (IITM), Chennai, India, in 2010.

From December 2009, he is with the Department of Mathematics, University of Coimbra, Portugal as a post-doctoral fellow. His research interests include mathematical image processing, anisotropic diffusion models, and variational and regularization methods for inverse problems.



Arindama Singh received his Ph.D. degree in mathematics from Indian Institute of Technology Kanpur (IITK), India, in 1990.

From 1991 to 1995, he was at the Department of Mathematics, University of Hyderabad as a assistant professor. In December 1995, he joined the Department of Mathematics, Indian Institute of Technology Madras, Chennai, India and is currently a full professor. His major areas of research are numerical analysis, knowledge compilation, mathematical logic, regularization, and PDE-based image processing.